

Michael J. Barnsley

ENVIRONMENTAL MODELING



A Practical Introduction



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Taylor & Francis Group
Boca Raton London New York

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CRC Press
Taylor & Francis Group
6000 Broken Sound Parkway NW, Suite 300
Boca Raton, FL 33487-2742

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Printed in the United States of America on acid-free paper
10 9 8 7 6 5 4 3 2 1

International Standard Book Number-10: 0-415-30054-1 (Hardcover)
International Standard Book Number-13: 978-0-415-30054-4 (Hardcover)
DOI: 10.1201/9781315275970

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Library of Congress Cataloging-in-Publication Data

Barnsley, Michael J. (Michael John)
Environmental modeling : a practical introduction / Michael John Barnsley.
p. cm.
Includes bibliographical references and index.
ISBN-13: 978-0-415-30054-4 (alk. paper)
ISBN-10: 0-415-30054-1 (alk. paper)
1. Environmental sciences--Mathematical models. I. Title.

GE45.M37B37 2007
577.01'5118--dc22

2006030555

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<http://www.crcpress.com>

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Preface

The initial motivation to write this book stemmed from the perception of a growing divide between the increasingly sophisticated computer-based models that are being developed to represent various aspects of Earth's environmental systems (including those pertaining to its climate, ecosystems, biogeochemical cycles and hydrological processes) and the ability of many undergraduate students, and even some graduate students, of the environmental sciences to engage constructively with these models. The aim of this book, therefore, is to provide a practical introduction to the various methods, techniques and skills involved in computerized environmental modeling, including (i) representing an environmental problem in conceptual terms (i.e., developing a conceptual model), (ii) formalizing the conceptual model using mathematical expressions (i.e., formulating a mathematical model), (iii) converting the mathematical model into a program that can be run on a desktop or a laptop computer (i.e., implementing a computational model) and (iv) examining the results produced by the computational model (i.e., visualizing the output from a model and checking the model's validity in comparison with observations of the target system).

The contents of this book are based on a course that I have taught for many years to honors degree undergraduate students at Swansea University. The objectives of the course, and of this book, are to introduce the student to the broad arena of environmental modeling, to show how computational models can be used to represent environmental systems, and to illustrate how such models can improve our understanding of the ways in which environmental systems function. Equally important, the book also aims to impart a set of associated analytical and practical skills, which will allow the reader to develop, implement and experiment with a range of computerized environmental models. The emphasis is, therefore, on active engagement in the modeling process rather than on passive learning about a suite of well-established models. A practical approach is adopted throughout, one that tries not to get bogged down in the details of the underlying mathematics and that encourages learning through "hands on" experimentation. To this end, a set of software tools and data sets are provided free-of-charge under the General Public License (GPL) and Gnuplot License so that the reader can work through the various examples and exercises presented in each chapter.

Most of the data sets used in this book relate to an area immediately south of Llyn Efyrynwy, Powys in mid-Wales, UK. Apart from the fact that this is a particularly beautiful part of the world, I chose this site because it is one of a relatively small

number of locations in the UK at which the solar irradiance measurements used in Chapter 5 are routinely recorded. I should also confess to deriving a certain amount of innocent amusement thinking about the additional challenge that the pronunciation of this particular Welsh place name will present to many readers. If nothing else, it will help to take the reader's mind off the demands of environmental modeling, every now and then.

I am deeply grateful to the UK Ordnance Survey, and in particular Ed Parson, for making available the digital elevation data used in Chapters 2 and 10. Thanks are also due to the UK MetOffice for permission to use the various meteorological data sets pertaining to Llyn Efyrynwy, and also to the staff at the British Atmospheric Data Centre (BADC), which is operated by the UK's Natural Environmental Research Council (NERC), for providing the excellent service through which I was able to access these data.

This book was put together using a range of “open source” software, including the GNU/Linux operating system. Most of the figures were produced using gnuplot (<http://www.gnuplot.info/>); the majority of the remainder were created using the PSTricks class in $\text{\LaTeX}$. $\text{\LaTeX}$ (<http://www.lyx.org/>) and $\text{\LaTeX}$ (<http://www.latex-project.org/>) were used to produce the camera-ready copy, and the Beamer class in $\text{\LaTeX}$ was used to create the presentation files. I should like to thank the developers of all of these software packages.

I should also acknowledge the many cohorts of undergraduate students at Swansea University who have acted as a test bed for much of the material presented in this book: a sea of blank faces is undoubtedly the most immediate and effective signal that the material being presented is inadequately explained or otherwise confusing, and I hope that the salutary lessons that my students have taught me along the way have resulted in a clearer exposition in this book. On a more positive note, I am deeply gratified by those students who, having been introduced to computer-based environmental modeling for the first time, have honed their newly acquired skills and gone on to greater things. I hope that this is also the case for the readers of this book.

Finally, I should like to acknowledge the support of various friends and colleagues who have offered help and advice, and above all provided much-needed injections of humor at numerous points during the production of this book: to Mat Disney, Philip Lewis (“Lewis”) and Tristan Quaife at University College London, to Tim Fearnside, Sietse Los, Adrian Luckman, Peter North and Rory Walsh at Swansea, and to Paul Mather at the University of Nottingham, *diolch yn fawr iawn* (thank you very much). Paul Mather deserves special mention for kindly reading through the drafts of this book and for providing many detailed comments and helpful suggestions. As it has become conventional to say at this point, though, any oversights, omissions or errors that remain are mine alone. I understand that in the world of “closed source” computer software such things are often described as “features”; I hope, however, that “features” of this type are few and far between herein. *Pob lwc!* (Good luck!)

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Chapter 1

Models and Modeling

Topics

- Why model?
- The modeling process
- A typology of models
- Systems analysis and systems dynamics

1.1 WHY MODEL?

A model is a simplified representation of a more complex phenomenon, process or system; an environmental model is one that pertains to a specific aspect of either the natural or the built environment. Environmental models have been developed to represent, among other things, elements of Earth's climate system, hydrological processes, ecosystems and biogeochemical cycles. The principal purposes of these models are threefold: to increase knowledge, and hence to reduce uncertainty, of the phenomenon, process or system that the model purports to represent (i.e., to improve understanding); to provide a tool with which to estimate the state of the phenomenon, process or system at times and locations other than those for which observations are presently available (i.e., to facilitate prediction); and to provide a framework within which “what if” questions can be asked about possible changes to the state and operation of the phenomenon, process or system under specified conditions (i.e., to perform simulations).

The production of an environmental model involves a process of abstraction: in the sense that most environmental models deal with abstract concepts and ideas (i.e., mathematical formulae and computational code) rather than physical objects

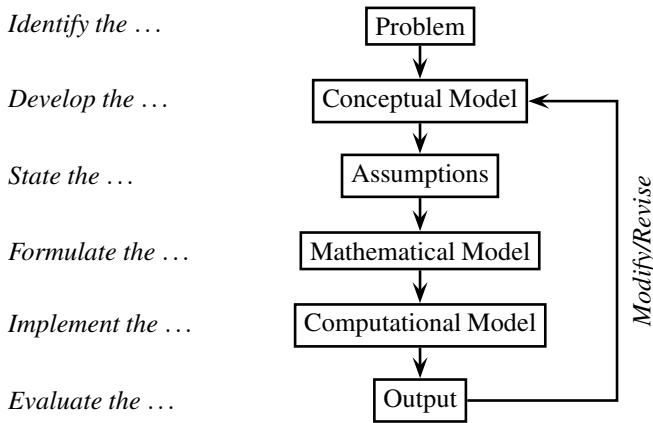


Figure 1.1: Schematic representation of the modeling process.

and events (i.e., conceptualization); in terms of identifying and extracting the most important elements of the phenomenon, process or system and discarding the least significant ones (i.e., selection); and in the sense of summarizing the essence of the phenomenon, process or system (i.e., encapsulation). Building a well-designed model therefore forces one to examine carefully, analytically and in detail the component elements of an environmental system, the processes and structures that govern the relationships and interactions between them, and the spatial and temporal scales over which they operate.

1.2 THE MODELING PROCESS

In principle, the process of designing, building and using an environmental model can be divided into a series of discrete stages. These stages are shown schematically in Figure 1.1 and are described in detail below. In practice, the boundaries between the different stages are not always well defined and progression from one stage to the next is seldom as straightforward or as linear as Figure 1.1 implies. Nevertheless, this diagram provides a useful framework within which to introduce the basic concepts.

1.2.1 Identifying the Nature and Scope of the Problem

The first step is to identify the specific science question, or problem, that is to be addressed and then to establish both whether and how a model will help to answer this question (Wainwright and Mulligan 2004). The problem should be sufficiently well defined and focused so that it is amenable to solution using the knowledge, skills and resources at hand (these factors influence the tractability of the problem), but it should also be sufficiently generic so that it is of more than just parochial interest (this implies that a compromise is negotiated between the specificity and the generality of the model). If the problem is poorly defined at the outset, the model-building process

Table 1.1: Four main phases of systems analysis (after Huggett 1980).

Phase	Actions
Lexical	Define the system boundaries (closure). Choose the system components, i.e., state variables (entitation). Estimate the values (i.e., the state) of the state variables (quantitation).
Parsing	Define verbally, statistically or analytically the relationships between the state variables.
Modeling	Model construction. Model operationalization (i.e., running the model).
Analysis	Model validation and verification (i.e., compare the results of the model with observations of the target system).

is likely to be more difficult, more time-consuming and more complex. Worse still, the resultant model may not be appropriate to the task for which it was originally intended.

A related consideration is the scope of the model, in terms of those elements of the science question that the model is, and is not, intended to address. The scope of the model may have to be limited in various ways to produce a tractable solution. For instance, the model may need to be designed so that it represents a selected part of the target environmental system, a particular spatial domain, a specified period of time, or perhaps a combination of all three.

1.2.2 Developing the Conceptual Model

After specifying the science question, the next step is to develop a conceptual model of the problem. The term *conceptual model* is used here to refer to a model that is expressed verbally or in written or diagrammatic form (i.e., concepts), as distinct from one that is represented in terms of mathematical formulae (i.e., a mathematical model) or one that is constructed from physical materials (i.e., a physical model).

The development of a conceptual model necessarily involves a comprehensive analysis of the target phenomenon, process or system with the aim of identifying its component parts, their respective inputs and outputs, the relationships between them, and the processes and structures that govern their interaction. This stage in the model-building process is therefore closely related to the lexical and parsing phases of systems analysis, a branch of science concerned with the study of complex systems, including their composition, structure, function and operation (Huggett 1980, Table 1.1). In each case, it is assumed that the “real world” can be divided into a number of more or less discrete systems, which can be further sub-divided into their component parts and processes, identified by careful analysis and detailed observation (Hardisty *et al.* 1993).

Table 1.2: Important definitions in environmental modeling.

Element	Definition	Example
Constant	Quantity whose value does not vary in the target system.	Speed of light.
Parameter	Quantity whose value is constant in the case considered, but may vary in different cases.	Total solar radiation at the top of Earth's atmosphere.
Variable	Quantity whose value may change freely in response to the functioning of the system.	Amount of precipitation.
Relation	Functional connection or correspondence between two or more system elements.	Rainfall, run-off and soil erosion.
Relationship	State of being related.	—
Process	Operation or event, operating over time (temporal process) or space (spatial process) or both, which changes a quantity in the target system.	Evapotranspiration.
Scale	Relative dimension, in space and time, over which processes operate and measurements are made.	Local, regional, global; diurnal, seasonal, annual.
Structure	Manner in which component parts of a system are organized.	—
System	Set of related elements (e.g., constants, parameters and variables), the relations between them, the functions or processes that govern these relations and the structure by which they are organized.	Forest ecosystem, drainage basin, global carbon cycle, Earth's climate.

The component elements of an environmental system typically include inputs, outputs, constants, parameters, variables (also known as stocks, stores, pools and reservoirs), processes (flows), relations (links or connectors) and structures (Edwards and Hamson 1989); see Table 1.2 for definitions. The boundaries, or limits, of the target environmental system must also be specified. In this context, environmental systems are sometimes classified in terms of their degree of openness: open systems, also known as forced systems, have exogenous (or forcing) variables; closed systems, also known as unforced systems, have no exogenous variables (i.e., all of the variables are endogenous to the system) (Hardisty *et al.* 1993).

The process of developing a conceptual model is often aided by using diagrams to represent the component parts of the target system and the connections between them (e.g., Figure 1.2). These diagrams vary from the simple to the complex, from the

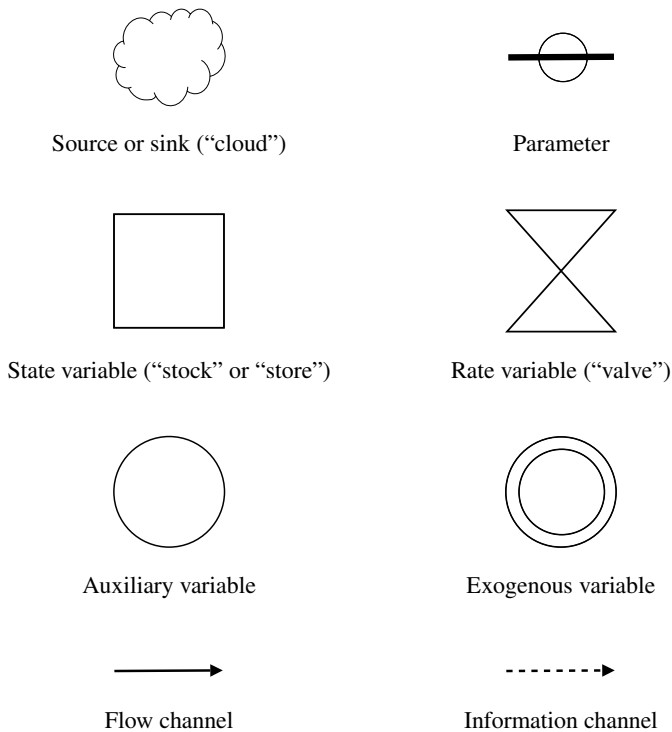


Figure 1.2: Examples of some of the symbols used in Forrester diagrams.

schematic to the formalized. The symbols employed for this purpose differ somewhat between studies, although Forrester diagrams (Forrester 1973) and their derivatives are used quite widely, particularly in the field of ecosystem dynamics (Ford 1999, Deaton and Winebrake 2000).

1.2.3 Stating the Assumptions

Every environmental model is founded on a set of assumptions. These assumptions may be made so that a complex environmental system can be simplified sufficiently to produce a working model (e.g., when deciding which elements of the target system should be included in the model and which should be omitted) or they may reflect the limits to current knowledge of the target environmental system (e.g., concerning the nature and form of the relations between its component parts). The validity and the scope of any such assumptions ultimately determine the value of the resultant model (Edwards and Hamson 1989, Wainwright and Mulligan 2004).

Some of the assumptions that are made when a model is first created may later be found to be incorrect, in which case it is often possible to revise the assumptions in subsequent versions of the model. Some may be known to be wrong at the outset, but nevertheless they may be retained because they have a relatively insignificant effect

on the output of the model and are necessary for reasons of simplicity or efficiency (Wainwright and Mulligan 2004). What is important is that each of the assumptions inherent in a model is recognized, understood and stated explicitly (Edwards and Hamson 1989, Wainwright and Mulligan 2004). The primary reasons for this are twofold: first, it clarifies the nature, purpose and limitations of the model, not least in the mind of the modeler; second, it helps potential users of the model to understand its scope (i.e., the range of conditions over which the model is known, or thought, to be valid), to challenge the assumptions on which it is based and, hence, to develop improved versions of the model in the future. For both the modeler and the user, it is particularly important to consider the consequences, whether they are intended or not, of the assumptions made in the model and to identify any assumptions that may have been made implicitly (i.e., without recognizing that this is the case) (Edwards and Hamson 1989).

Although it is not always mentioned in this context, the modeler should make clear the spatial and temporal scales over which the relations and processes that are being modeled operate. Many environmental processes operate, or are dominant, over a restricted range of spatial and temporal scales, and different processes operate at different scales; this phenomenon is sometimes referred to as “domains of scale” in the field of landscape ecology (Wiens 1989). It is inadvisable, therefore, to apply a model outside the range of scales for which it is designed; that is to say, most models are scale dependent.

1.2.4 Formulating the Mathematical Model

The next stage is to represent the conceptual model in mathematical terms; that is, using mathematical tools and concepts, such as variables, functions and equations. This process can be described as one of formulating the mathematical model, since it involves translating the conceptual model into mathematical formulae (Edwards and Hamson 1989). It is frequently the most challenging stage in the development of a model. Sometimes this is because the solution to the problem demands the use of advanced mathematical techniques, although each of the models considered in this book requires only basic skills in algebra and trigonometry. More often it is because there is more than one way in which the system can be represented mathematically and it is not immediately apparent which approach is best. Therefore, deriving a suitable mathematical formulation of a model is often a trial-and-error process, but it is also a skill that improves with practice.

The range and the diversity of mathematical models that have been developed to study environmental systems are considerable, and various different schemes have been proposed to group them by type. While these classification schemes differ at the level of detail, most are founded on a common set of principles that include the following considerations: the extent to which the model is derived from theory or from observations (i.e., empirical models versus theoretically informed models); the degree to which random events and effects play a major role in the target system and, hence, in the model (i.e., deterministic versus stochastic, or probabilistic, models); the level of knowledge or understanding of the target system that the model purports

to represent (i.e., black box versus white box models); whether the model is operated predominantly in forward or inverse mode (i.e., forward-mode versus inverse-mode models); whether the model deals with environmental processes that are static or dynamic with respect to time and space (i.e., static versus dynamic models); whether the model deals with environmental processes that can be considered to operate in a discrete or continuous manner with respect to time and space (i.e., discrete versus continuous models); and whether the model parameters are lumped or distributed (i.e., lumped-parameter versus distributed-parameter models). Each of these considerations is examined in greater detail in the sections that follow.

Empirical and Theoretical Models

An empirical model is one that is primarily or solely based on observations, as distinct from one that is derived from theory. In models of this type, the relationships between the component elements of an environmental system are established by examining measurements of the variables concerned. The form of each of these relationships is then defined by a mathematical function (Edwards and Hamson 1989). The decision as to which one of a set of candidate mathematical functions (e.g., Figure 1.3) should be used typically involves a compromise between how well the candidate functions fit the data and the relative simplicity of their mathematical form; the decision, however, is not informed by theory (Edwards and Hamson 1989). Regression analysis is widely used in this context to fit the function to the data.

While empirical models are frequently valuable in terms of making predictions in the cases for which they are developed, they typically lack generality (i.e., they are specific to a particular circumstance or set of data). It is often difficult, therefore, to employ them at other spatial locations or different points in time.

Deterministic and Stochastic (Probabilistic) Models

Loosely defined, a deterministic model is one in which the outputs (i.e., the results) are uniquely and consistently determined by the inputs (i.e., the values used to drive the model) (Edwards and Hamson 1989). Deterministic models therefore function in the same way, and produce exactly the same output, each time they are run using a particular set of input values. This definition can be recast a little more precisely as follows: the state (i.e., the value) of a variable in a deterministic model is uniquely and consistently determined by the initial conditions of the model (i.e., the values of the constants and parameters input to the model) and, subsequently, the previous states of the variable itself. In contrast to empirical models, the implication is that deterministic models are based on assumptions, theory or knowledge of the nature and form of the relations between the variables in the target system. Deterministic models tend, therefore, to offer a much greater degree of generality than empirical models. Consequently, if they are properly configured and suitably implemented, deterministic models can usually be applied to different spatial locations and points in time from those for which they were originally developed and tested.

In contrast, stochastic (or probabilistic) models are ones in which random events and effects play an important role (Edwards and Hamson 1989). The state of the

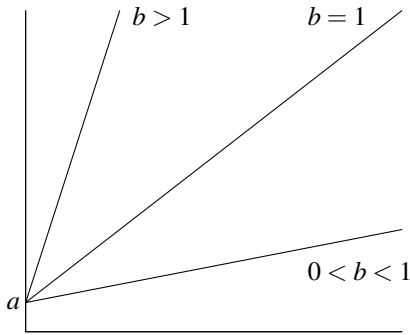
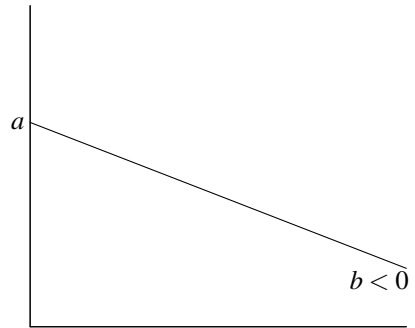
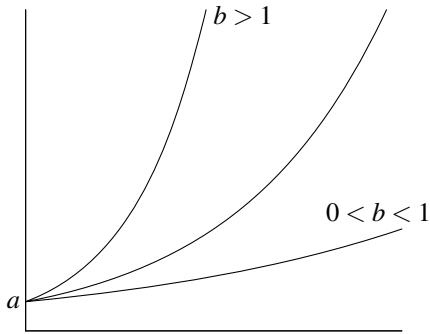
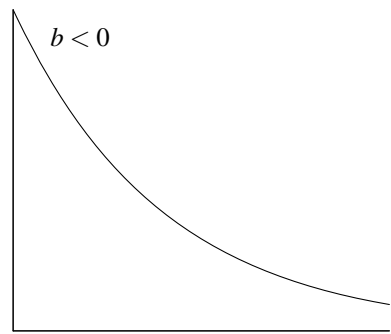
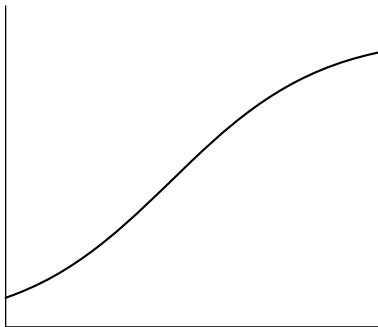
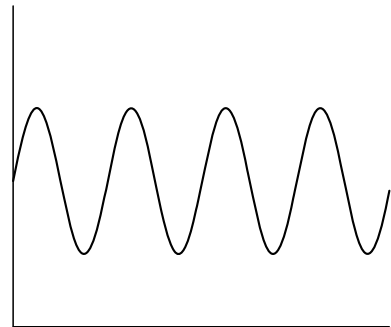
a) Positive linear, $y = a + bx$ b) Negative linear, $y = a + bx$ c) Positive exponential, $y = a \exp(bx)$ d) Negative exponential, $y = a \exp(bx)$ e) Logistic, $y = \frac{K}{\left(1 + \left(\frac{K - y_0}{y_0}\right) \exp(-rx)\right)}$ f) Periodic, $y = \sin(x)$

Figure 1.3: Forms of various mathematical functions.

model variables is therefore described by probability distributions, rather than single values. As a result, the output from such a model will vary from run to run even when the input values used are the same. This class of model is therefore suitable in those circumstances where apparently random fluctuations in the system processes, and hence in the system variables, render deterministic models inappropriate (Kirkby *et al.* 1993). The random fluctuations that are observed in the system, and hence in the model, may be due to environmental processes and events that are truly random in nature or they may be pseudo-random; that is, where knowledge of a potentially deterministic process is inadequate or incomplete, such that it has to be treated as though it is random.

Black Box and White Box Models

Mathematical models can also be classified according to the degree to which the composition, structure and operation of the target system is known and, hence, is represented in the model. The two extreme cases in this respect are usually referred to as white box (or clear box) models and black box models. In white box models, the internal workings of the target system are known, completely understood and clearly stated. This *a priori* information may be based on detailed observation of the target system or it may be derived from theory (Kirkby *et al.* 1993). By contrast, in black box models the target system is treated as a sealed unit, with no attempt made to understand the variables of which it is composed or the relations between them. Whereas white box models are based on knowledge or theory of the target system, black box models are usually defined empirically (Kirkby *et al.* 1993). In practice, most models fall somewhere between these two extremes; as such, they might be described as gray box models.

Forward and Inverse Modeling

Mathematical models are typically specified in terms of functional relations between two or more variables of the target system. For example, all other things being equal, the growth and abundance of a particular plant species in a given environment is functionally related to the ambient light conditions, the air temperature, the amount of precipitation and the availability of nutrients in the soil. A functional relation, such as this, can be expressed mathematically as follows:

$$growth = f(light, temperature, precipitation, nutrients) \quad (1.1)$$

Equation 1.1 indicates that the variable *growth* is a function *f* of four other system variables: *light*, *temperature*, *precipitation*, and *nutrients*. Thus, the value of *growth* is related to, and is in some way dependent on, the values of these four variables. If the form of the function *f* is known, or can be derived from theory or else be obtained by induction from a set of observations (i.e., empirically), it can be expressed as a rule, which allows the value of the dependent variable (i.e., *growth*) to be determined for any value of the independent variables (i.e., *light*, *temperature*, *precipitation*, and

nutrients). For instance, if $y = f(x)$ describes a functional relation between two system variables x and y , and the function f is defined in terms of the rule $f(x) = x^2$, then $y = f(3) = 3^2 = 9$ when $x = 3$. This will be referred to as forward modeling, for reasons that will become clear once inverse modeling is introduced, below.

Sometimes the modeler may be more interested in the inverse functional relation between two or more variables in the target environmental system. For instance, the modeler may wish to use knowledge of the relation between temperature and plant growth to reconstruct past climatic conditions (e.g., temperature) from an analysis of the abundance of a given plant species (e.g., inferred from pollen counts in cores taken from lake sediments). Inverse functional relations, such as this, can be expressed mathematically as follows:

$$x = f^{-1}(y) \quad (1.2)$$

Equation 1.2 states that there is a function f^{-1} , which is the inverse of function f , that relates the value of variable y to that of variable x (Biggs 1989, Piff 1992). Note that this does not imply that x is dependent on y (e.g., ambient temperature is not controlled in a direct way by the abundance of a particular plant species), merely that they are functionally related in some way. The rule for f^{-1} is obtained by reversing the rule for f (Grossman 1995). For example, if $y = f(x)$ and $f(x) = x^2$, then $x = f^{-1}(y)$ where $f^{-1}(y) = \sqrt{y}$. This is referred to as the inverse model. For simple empirical models, obtaining the rule for the inverse function is relatively straightforward: the values of variable y are regressed on the corresponding values of variable x , rather than *vice versa*. For more complex deterministic and stochastic models, however, the problem is much more challenging and involves the use of analytical and numerical solutions that are beyond the scope of this book.

One area in which the application of inverse models is increasingly common is the field of Earth observation, also known as terrestrial remote sensing. In part, this area of science is concerned with the reflectivity, at different wavelengths, of Earth's surface materials (e.g., vegetation, soil, rocks, snow, ice and water). The reflectivity of these materials is known to be a function of their respective physical, chemical and biological properties. Various models have been developed to represent these functional relations. These models can be used to predict the amount of solar radiation (i.e., sunlight) reflected by a terrestrial surface, based on information about its physical, chemical and biological properties. The inverse functional relation, however, is typically of greater interest for practical purposes; that is, the ability to estimate the properties of Earth's surface materials from measurements of spectral reflectance made by sensors mounted onboard aircraft or Earth-orbiting satellites. This requires the use of sophisticated analytical and numerical techniques to invert the forward model against the measured data (Goel 1989, Kuusk 1995, Verstraete *et al.* 1996).

Static Models, Dynamic Models, Equilibrium, Stability and Feedback

Mathematical models can be divided into two further categories: static and dynamic (Ford 1999). The former deal with systems that do not change, or at least are thought not to change, appreciably with respect to time. This type of model focuses on the

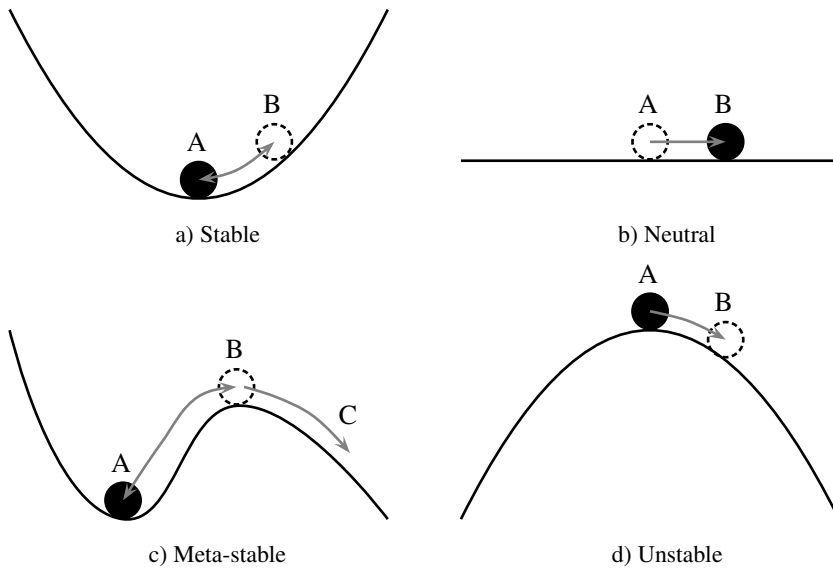


Figure 1.4: Graphical representation of different system states. The system state is represented by the position of a ball on the topographical surface.

processes, or forces, that keep the system in a state of equilibrium. Dynamic models, in contrast, deal with systems that change over time, which is much more common in environmental systems. Dynamic mathematical models are typically constructed from difference equations or differential equations (see Chapter 8).

The need to understand the behavior of an environmental system with respect to time leads the modeler to consider the important issue of system stability. A number of generic system states can be envisaged in this context; these states are known as stable, neutral, meta-stable and unstable (Huggett 1980, Figure 1.4). In a stable system, the system returns to its original state (position A, Figure 1.4a) having been perturbed (i.e., moved to position B, Figure 1.4a) by an external force or process. In a neutral system (Figure 1.4b), the action of an external force or process causes no change in the state of the system variables. In a meta-stable system, the system is, initially, in a weak stable state (position A, Figure 1.4c), but the effect of an external force or process may be to move it to an unstable transitional state (position B, Figure 1.4c) from which it may either return to its original state or change to a different one (position C, Figure 1.4c). Finally, an unstable system is one in which the system state changes, perhaps irrevocably, as a result of an external force or process (Figure 1.4d).

The relative stability of an environmental system is partly controlled by feedback mechanisms. Feedback is the process by which a fraction of the output from a given system, or part thereof (i.e., a sub-system), is returned (i.e., is fed back) as input to the same system or sub-system. This process is represented schematically in Figure 1.5. Feedback is an important feature of many environmental systems, not least because

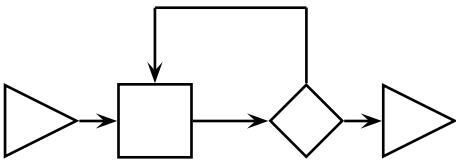


Figure 1.5: Schematic representation of a feedback relation.

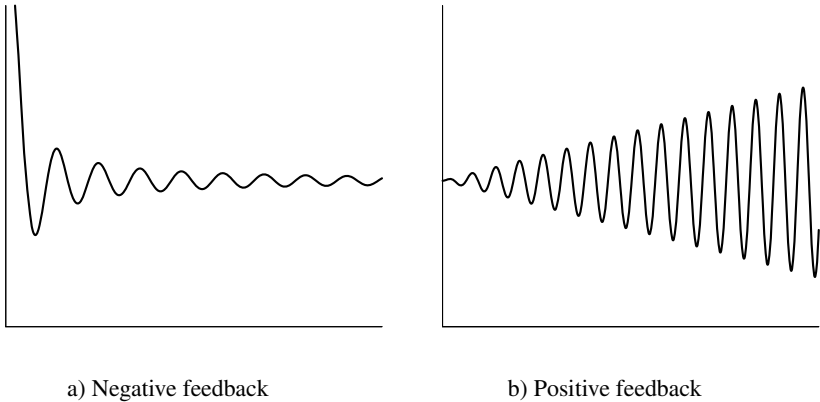


Figure 1.6: Negative and positive feedback relations.

it makes their operation more complex and their behavior more difficult to predict. Feedback is said to be either negative or positive (Figure 1.6). Negative feedback is produced by deviation-damping processes that counteract the effect of a perturbation to a system and, hence, tend to maintain the stability of the system: a system that displays a propensity for this type of behavior is said to exhibit homeostasis. In contrast, positive feedback is the result of processes that amplify an initial perturbation to a system and that tend to keep the system changing toward a new state: this tendency is sometimes referred to as homeorhesis. A notable example of a positive feedback loop in Earth’s climate system is one that involves sea ice and albedo (i.e., the average reflectivity of a surface). In this feedback loop, warmer atmospheric conditions (the initial perturbation) result in increased melting of the Arctic and Antarctic ice sheets. This reveals more of the relatively dark (i.e., lower albedo) polar oceans. These, in turn, absorb more solar radiation, which causes the oceans to warm further, so that still more of the polar ice sheets melt, and so on.

Before proceeding further, it is worth noting that many environmental systems not only exhibit feedback mechanisms but also branching or splitting events (Figure 1.7). These occur where two or more inputs feed into a single process, or where two or more outcomes are possible depending on certain conditions (Huggett 1980). As is the case with feedback mechanisms, branching events make the behavior of the system more complex and, hence, more difficult to predict.

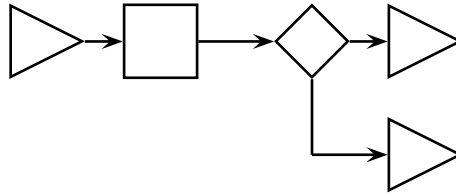


Figure 1.7: Schematic representation of a branching or splitting relation.

Feedback mechanisms and branching events are examined in greater detail in the context of the Daisyworld model, a simple biospheric feedback model, which is analyzed in Chapter 9.

Continuous and Discrete Models

Mathematical models can also be divided into two further categories: discrete and continuous. The distinction between the two relates to the way in which each treats time (or spatial location or both) as a variable. In discrete models, time proceeds in a series of finite steps that are usually, although not necessarily, of equal length; each step relates to a specific interval, such as an hour, a day, a month or a year. The state of the system is updated at the end of each time step, taking into account changes in the system variables resulting from the processes operating on the system over that time step. In continuous models, by contrast, the system state is updated instantaneously in response to continuous changes in the forcing factors. Discrete and continuous models also differ in terms of the mathematical equations on which they are founded: the former are expressed in terms of difference equations; the latter employ differential equations. Discrete and continuous mathematical models are explored in detail, in the context of studies of population growth, in Chapter 8.

Distributed-Parameter and Lumped-Parameter Models

Environmental systems often exhibit considerable spatial heterogeneity in the value of their variables and in the processes that control them. Some environmental models account for this spatial variation explicitly in their mathematical formulation. This is achieved by dividing the geographical domain of the model into a number of discrete spatial units, such as tessellating polygons (particularly square cells, which are then known as rasters), triangulated irregular networks (TINs) or irregularly shaped spatial objects (Wainwright and Mulligan 2004). Models that assign different values of the system parameters and variables to each of these spatial units are known as distributed-parameter models. This approach is not always feasible, however. Sometimes this is because of computational constraints; on other occasions it may be due to a lack of data on the spatial variability of the model parameters and variables. In such circumstances, it may be necessary to assign a single “lumped” value across the whole of the model’s spatial domain; models such as this are often referred to as lumped-parameter models.

Analytical and Numerical Solutions to Mathematical Models

It is possible, in many cases, to construct a mathematical model of an environmental system by analytical means; that is, by formulating the model so that it is expressed concisely in terms of mathematical equations, functions, variables and constants. This approach is sometimes referred to as an analytical or closed-form solution. Not all environmental models are amenable to this approach, however; some must be solved numerically. The numerical approach relies heavily on the power of modern computers to solve the model by performing a sequence of operations rapidly over and over again, each time applying the operations to the result of the previous iteration (British Computer Society 1998). Sometimes an estimate, or guess, is made regarding the correct solution, to which the iterative procedure is initially applied. Provided that the mathematical model has been formulated and implemented correctly, the expectation is that the procedure will move ever closer to the correct solution with each successive iteration. Depending on the nature of the model, and in some cases the accuracy of the initial estimate, the iterative procedure will eventually reach the correct solution or else a very close approximation to it. The best known example of this approach is Newton's iterative method for calculating $\sqrt{x}$ (Harris and Stocker 1998). Examples of the analytical and numerical approaches to modeling are examined in detail in Chapter 7.

1.2.5 Implementing the Computational Model

General Considerations

Once an appropriate formulation has been derived for the mathematical model, the next step is to convert the equations and formulae into some form of software that can be run on a computer; that is, to implement the computational model. There is an enormous range of options available to the modeler in this context, including spreadsheet packages, specialized modeling software, high-level computer programming and scripting languages, and integrated modeling environments (Table 1.3). Each has its advantages and disadvantages. Ultimately, the decision regarding which software the modeler chooses to use is likely to be conditioned by issues of availability, cost, personal preference and prior experience, in addition to the ease with which the software is learned and its suitability for the task at hand.

It has been argued that it is easier to learn how to use spreadsheet packages than formal programming languages (Hardisty *et al.* 1993). Spreadsheets are certainly ubiquitous and, hence, are familiar to many computer users. Most spreadsheet packages offer the added advantage of built-in facilities for data visualization, statistical analysis and function-fitting. Spreadsheets have been used quite widely, therefore, in environmental modeling; for example, in conservation biology and landscape ecology (Donovan and Welden 2001, 2002). Nevertheless, spreadsheets have intrinsic limitations that present problems when producing more sophisticated environmental models, especially those requiring iteration to reach the solution. Most importantly, perhaps, spreadsheets do not encourage a structured approach to model building, and the potential for sharing and reusing code is limited.

Table 1.3: Selection of tools suitable for implementing computer-based environmental models.

Category	Name	Reference
Spreadsheets	Calc	http://www.openoffice.org/
	Excel	http://office.microsoft.com/
	Gnumeric	http://www.gnome.org/projects/
	Kspread	http://www.koffice.org/
	Quattro Pro	http://www.corel.com/
Specialized modeling environments	MODELMAKER	http://www.modelkinetix.com/
	POWERSIM	http://www.powersim.com/
	SIMILE	http://simulistics.com/
	STELLA	http://www.iseesystems.com/
	VENSIM	http://www.vensim.com/
High-level computer programming languages	BASIC	http://www.freebasic.net/
	C	Kernighan and Ritchie (1988)
	C++	Oualline (1995)
	FORTRAN	Hahn (1994)
	JAVA	Niemeyer and Knudsen (2005)
	PASCAL	Buchanan (1989)
	VB	Willis and Newsome (2005)
Scripting languages	awk	Aho <i>et al.</i> (1988)
	JavaScript	Flanagan (2001)
	Perl	Wall and Schwartz (1993)
	Python	Lutz (1996)
	PHP	Atkinson (2000)
	Rexx	Cowlshaw (1990)
	Ruby	Thomas and Hunt (2000)
	Tcl/Tk	Welch <i>et al.</i> (2003)
Integrated modeling environments	IDL	http://www.itvis.com/
	MATLAB	http://www.mathworks.com/
	OCTAVE	http://www.gnu.org/software/octave/
	R	http://www.r-project.org/
	S	http://www.insightful.com/

Specialized modeling software also offers many important benefits, including the ability to construct models diagrammatically using graphical interfaces, as opposed to typing lines of computer code, and the provision of an extensive range of built-in analytical and numerical functions and data visualization routines. This type of software tends to be made available on a commercial basis, however; that is, at some financial cost to the modeler. It may also be difficult to share models developed in one software package with users who employ a different modeling suite. As a result, there is a danger that the modeler becomes “locked in” to a product offered by a particular vendor. Finally, the modeling software may restrict the user to a particular set of modeling techniques, depending on the functions and the range of expressions offered by the software.

High-level computer programming and scripting languages represent yet another option for the modeler. In many senses this is the most flexible option, albeit one that requires the modeler to learn the commands, grammar and syntax of the chosen language. Despite suggestions to the contrary, however, this is not a major challenge and, importantly, the reward is a highly transferable skill that can be applied in a wide range of contexts. The choice of which programming language to use is largely a matter of personal preference. Although there are differences between programming languages, most can be used effectively to implement a computer-based environmental model. Moreover, many programming languages share a set of common features so that once the user has learned one language it is relatively easy to master another.

Irrespective of the specific software that is used for the purpose, the process of implementing a computational model involves a translation between the vocabulary, grammar and syntax of one language (i.e., the functions and equations of mathematics) and those of another (i.e., the instructions of a spreadsheet or high-level computer programming language). Just as in the translation between natural languages, such as English and French, there is rarely a single “correct” solution to this process, although some solutions might be regarded as being intrinsically better than others. The flexibility and versatility of computer software means that the implementation can often be realized in a number of ways. In some instances, the differences may amount to little more than a matter of programming style, broadly analogous to the variations in style of written prose among literary authors; on other occasions, however, the differences may have a significant impact in terms of the efficiency of the resulting code, including the speed with which the result is computed and the demands that the program places on various aspects of the computer’s resources, such as memory usage and hard-disk access. Thus, some implementations may be considered to be more elegant or otherwise preferable to others.

Computer Software Used in This Book

A simple, but powerful, high-level scripting language, known as *awk*, is used in this book. The acronym *awk* is derived from the surnames of the individuals who initially developed the *awk* language and utility: Aho, Weinberger and Kernighan (Aho *et al.* 1988). There are several versions of *awk*. The version used here is the one produced by the GNU Project (<http://www.gnu.org/>), which is known as *gawk*.

(Dougherty 1996, Robbins 2001). *gawk* has two main advantages in this context: it is an interpreted language, meaning that the computer interprets each line of the program (or script) as it runs, so that the modeler does not have to compile the code before it is executed by the computer; and it is data-driven, in as much as it provides a simple mechanism for reading data from files structured in terms of fields (columns) and records (lines or rows). The second of these two features greatly simplifies the process of incorporating data to models. More generally, *gawk* provides a powerful and flexible tool for data manipulation and a convenient framework within which to develop a range of environmental simulation models. It is also available free-of-charge for use on a wide range of computer platforms and operating systems.

While the *gawk* programming language is used to manipulate data stored in files and to implement computational models, a separate utility is required to visualize the inputs to, and outputs from, models. A software package known as *gnuplot* is used for this purpose. *gnuplot* is primarily, but not exclusively, an interactive, command-driven, function and data plotting program (Williams and Kelly 1998). It can be used to generate two-dimensional (2D) scatterplots and line diagrams in either rectangular or polar coordinates, as well as plots of points, lines, vectors and surfaces in three dimensions (3D). It is able to fit user-defined functions to a data set and to output the results to a wide range of graphical file formats, as well as to the computer screen and printer. Apart from its analytical and plotting capabilities, *gnuplot* has two other important advantages: it is available free-of-charge under the terms of its license (Appendix C) and versions of the software are available for use on a wide variety of computer platforms and operating systems.

Model Parameterization

Before proceeding further, it is worth noting that many computational models require what is sometimes known as parameterization (also referred to as parametrization or parameter estimation); that is, the values of the parameters and the initial values of the variables used in the model need to be determined so that the model produces sensible results (i.e., the model output matches observations of the target system under the conditions being modeled). This process is also known as tuning or calibrating the model. Parameterization can be performed graphically (i.e., visually), statistically (e.g., using least-squares estimation, often known as regression) or analytically (e.g., by solving a set of simultaneous linear equations).

1.2.6 Evaluating the Model

Verification and Validation

Once the computational model has been implemented, the next step is to run the model to check, first, that it works and, second, that it produces acceptable results. If the result of the first of these tests is unsatisfactory, the implementation should be checked and, where appropriate, revised. The second test is addressed by a process known as model verification and validation (or V&V). The difference between *verification* and *validation* is a subtle one: verification involves an assessment that the